

Plan: "Food Facilities" call-list tab for Zest Commercial Leads

Status: **plan only, nothing built yet.** Revised 2026-09-08 (v2) after owner input: 28-mile radius from the office, category framing, and the question "is there a permit or portal that beats Google?" (answered in section 3). No code has been written and the Google Sheet has not been touched.

Target sheet: *Zest Commercial Leads* (<https://docs.google.com/spreadsheets/d/1u6e6psiW581rGr92r86aSJQW2XzPZfzsnhpJl0lB4F8>).

1. Goal

Add one new tab that a salesperson works top-to-bottom to sell pest-control inspections to **food processing facilities, food manufacturers, food storage facilities and other wholesale food facilities** within 28 miles of Zest's office. Restaurants and retail markets are out; the county health-inspection pipeline in the **Leads** tab already covers them. Every row needs business name, phone, type of business and city, plus Notes and Follow-up date for the rep.

2. What is in the sheet today (read-only look)

TAB	SHAPE	WHAT IT IS
Leads	205 rows x 41 cols	County health-inspection leads (Sacramento EMD, Placer, Yolo), keyed by permit ID. Rep-owned columns already exist: status, notes, rep, last touch date, next step date, touch count, inspection date, outcome, FieldRoutes customer ID.
Signals	63 rows	One row per inspection report (vermin quote, result, report URL).
Runs	5 rows	Run log of the existing pipeline (rows added / flagged / skipped dnc / skipped cap / errors).
DNC	header only	Do-not-call list: key, phone, email, reason, added at.

No formulas, validation, conditional formatting or protected ranges anywhere. The existing pipeline writes by column header name, caps itself at 40 new rows per run and respects the DNC tab. The new tab follows the same conventions.

Why food processors are missing from Leads. County inspection portals cover *retail* food only. Wholesale processors, manufacturers and food warehouses are permitted by the state (CDPH Food and Drug Branch), meat and poultry by USDA FSIS or CDFA, dairies by CDFA. That is a different set of permits, covered next.

3. Is there a permit or portal? Yes, several, but no single one (owner's question)

Checked 2026-09-08. Short answer: the one permit that covers exactly this population is **not published online**, but a handful of free permit and registry sources exist and should be pulled *before* Google. Google then fills the gaps and supplies phones for the sources that lack them.

SOURCE

CDPH Food and Drug Branch, Processed Food Registration (PFR)

City of Sacramento Business Operation Tax (BOT) open data

USDA FSIS Meat, Poultry and Egg Product Inspection Directory

EPA ECHO facility search

CalEPA Regulated Site Portal / CERS

CDFA Milk and Dairy Food Safety Branch

Data Axle Reference Solutions (free with a Sacramento Public Library card)

County health departments (Sacramento EMD, Placer, Yolo)

Other cities' business licenses (West Sacramento, Elk Grove, Roseville, Woodland, Davis, Folsom, Citrus Heights, Rancho Cordova)

Recommendation: layer them.

1. Pull the registries that carry phones (City of Sacramento BOT, FSIS) and the ones that do not (ECHO, CalEPA, CDFA). Merge on normalized name + address. This is the "permit-grade" core.
2. File the CDPH PRA request in parallel (I draft it; the owner sends it from a Zest address). When the list arrives, merge it in the same way.
3. Run the Google Places keyword sweep (section 4) to catch facilities no registry has, especially outside the City of Sacramento, and to add phone, website, Maps link and open/closed status to every row that lacks them.
4. If the owner has a library card, a Data Axle export becomes a fourth input and a completeness check.

Every row records its `Source(s)` so the rep and the owner can see where a lead came from.

4. Google Places API (New): gap-filler and enrichment

4.1 Why Text Search, and why keywords

Verified against Google's place-type table: **no place type exists for food processing, factory, warehouse, distribution center or cold storage.** Nearest: `manufacturer`, `supplier`, `whole saler`, `farm`, `bakery`, `butcher_shop`. So Nearby Search (type-driven) cannot find these; Text Search (keyword-driven) can. Text Search facts that shape the design: 20 results per page,

60 per query max (3 pages), `locationRestriction` accepts a **rectangle only**, one optional `includedType` with `strictTypeFiltering`.

4.2 Tiling

Each keyword runs first against the whole coverage rectangle. If it returns 60 (saturated), the rectangle is split into four quadrants and re-run, recursively, until no quadrant saturates. Results de-duplicate by place ID.

4.3 Search terms (draft of about 50, tuned after the first dry run)

Processing/manufacturing: food processing plant, food processing facility, food manufacturer, food manufacturing, food packaging company, co-packer, commercial bakery, wholesale bakery, tortilla factory, meat processing, meat packing, slaughterhouse, poultry processing, seafood processor, dairy processing plant, creamery, cheese manufacturer, egg processing, produce packing, fruit packing house, nut processing, almond processor, rice mill, flour mill, feed mill, cannery, frozen food manufacturer, snack food manufacturer, candy manufacturer, spice manufacturer, sauce manufacturer, pet food manufacturer, ice manufacturer.

Beverage: beverage manufacturer, bottling plant, juice processing, coffee roaster wholesale, brewery production facility, winery production facility, distillery.

Storage/distribution: food storage facility, cold storage warehouse, refrigerated warehouse, food distribution center, food distributor, wholesale grocer, produce distributor, meat distributor, seafood distributor, beverage distributor, food warehouse, food bank warehouse.

Kitchens: commissary kitchen, central kitchen, catering commissary, school district central kitchen.

Plus type-driven passes: `includedType = manufacturer, wholesaler, supplier` with query "food", strict filtering.

4.4 Enrichment for registry rows

For every registry row missing a phone or website, one Text Search by "name + address" (Pro fields) followed by a Place Details call (Enterprise fields: `nationalPhoneNumber`, `websiteUri`) fills the gaps and attaches the place ID.

5. Coverage area: 28 miles from the office

Center: **6948 West 2nd St, Rio Linda, CA 95673** (geocoded 38.6941, -121.4660). A candidate is kept only if its straight-line distance to that point is **28.0 miles or less**; the distance goes in its own column. Google searches use the 56 x 56 mile bounding rectangle, then rows are trimmed by the distance rule so the circle, not the square, is what ends up in the sheet.

In: Sacramento, West Sacramento, Woodland, Davis, Elk Grove, Rancho Cordova, Folsom, El Dorado Hills, Citrus Heights, Roseville, Rocklin, Lincoln, Wheatland, and Auburn at the edge.
Out: Galt, Dixon, Lodi/Stockton, Yuba City/Marysville, Placerville.

6. The new tab

Name: **Food Facilities** , appended as the last tab. Columns left to right in the order the owner asked for:

COL	HEADER	OWNER	NOTES
A	Business name	script	Cleaned name
B	Phone	script	(916) 555-1234 ; blank if no source has one
C	Type of business	script	Our plain-English category (section 7)
D	City	script	
E	Notes	rep	Free text. Never touched by the script
F	Follow-up date	rep	Date-validated. Never touched by the script
G	Status	rep	Dropdown: New, No answer, Callback, Interested, Inspection booked, Not interested, Bad number, Do not call
H	Rep	rep	
I	Last call date	rep	
J	Address	script	
K	Zip	script	
L	Website	script	
M	Google Maps link	script	
N	Open/closed	script	Google <code>businessStatus</code> , or registry status
O	Distance (mi)	script	From the office, straight line
P	Region	script	Same labels as <code>Leads</code>
Q	Source(s)	script	e.g. <code>City BOT</code> ; <code>FSIS</code> ; <code>Google</code>
R	Found via	script	Search terms or registry description that surfaced it
S	Google category	script	Raw <code>primaryType</code>
T	Needs review?	script	<code>yes</code> when classification is uncertain
U	In Leads tab?	script	<code>yes</code> on a name+zip or phone match against <code>Leads</code>
V	FieldRoutes customer ID	rep	Same convention as <code>Leads</code>
W	Key	script	Google place ID when known, else the registry record ID (BOT account number, FSIS establishment number). Do not edit
X	First seen	script	
Y	Last refreshed	script	

One-time formatting: frozen bold header, column widths, header filter, status dropdown, date validation on F and I, warning-only protection on script-owned columns, rows sorted by

distance. A "Source: Google" attribution in the header note where Google data is shown.

Rep-vs-script contract: the script writes only its own columns. On refresh it appends new rows, updates Open/closed and Last refreshed, and fills blank Phone/Website. It never rewrites a non-blank cell in A-D and never touches E-I or V.

7. Filtering and classification

Drop a candidate if its Google types include any restaurant-family type (`restaurant` , `*_restaurant` , `cafe` , `coffee_shop` , `bar` , `meal_takeaway` , `meal_delivery` , `food_court` , `ice_cream_shop` , `donut_shop` , `juice_shop` , `dessert_shop` , `sandwich_shop`) or retail type (`grocery_store` , `supermarket` , `convenience_store` , `liquor_store` , `asian_grocery_store` , `market` , `warehouse_store` , `candy_store` , `chocolate_shop`), or its name hits a blacklist (Restaurant, Cafe, Grill, Taqueria, Pizza, Sushi, Deli, Bistro, ...). Registry rows are filtered on their own description/NAICS instead (keep 311, 312, 4244, 4931; drop 722 food service and 445 food retail).

Keep `bakery` , `butcher_shop` , brewery and winery hits only when no restaurant/cafe type is present, flagged `Needs review? = yes` .

Type of business categories: Meat & poultry processing; Seafood processing; Dairy / creamery; Commercial bakery / tortilla; Produce packing / nut & rice processing; Snack, candy & confectionery manufacturing; Sauce, spice & prepared foods manufacturing; Frozen & ready-meal manufacturing; Beverage production; Cold storage / refrigerated warehouse; Food distribution / wholesale grocer; Commissary / central kitchen; Pet & animal food manufacturing; Food packaging / co-packer; Other food manufacturing. Assigned by registry NAICS/description first, then name keywords, then the search term, then Google `primaryType` .

8. How the other tabs stay untouched

- A Google Cloud **service account** is shared on this one spreadsheet as Editor; nothing else in Drive.
- Every write targets the `Food Facilities` sheet ID, resolved from the tab title and re-checked before each write. No spreadsheet-wide operations: no clears, no reordering, no column inserts elsewhere, no changes to `Leads` , `Signals` , `Runs` or `DNC` .
- Tab created with `addSheet` only if absent, appended at the end.
- Rows **upserted by Key** (column W); existing rows updated cell-by-cell in script-owned columns only.
- Before and after every run the script hashes the values of every *other* tab and refuses to report success if any changed.
- First run is a **dry run** to CSV; only after sign-off does `--write` run.
- Run log printed and saved locally, not appended to the other pipeline's `Runs` tab.

9. Cost, quota, time

Registries: free. Google (verified on Google's pricing page 2026-09-08, first tier):

SKU	PRICE / 1,000	FREE PER MONTH	OUR USE
Text Search Pro	\$32.00	5,000	Keyword sweep + name lookups: est. 500-1,200 calls
Place Details Enterprise (phone + website)	about \$20 (confirm)	1,000	One per kept row lacking a phone: est. 300-700
Text Search Enterprise	\$35.00	1,000	Not used

Expected: **\$0 to a few dollars** for the first pull, \$0 for a monthly refresh. A Google Cloud project with billing enabled is still required. Expected volume: roughly 400-800 facilities inside 28 miles. Full pull 10-20 minutes; raw responses cached to disk so re-runs while tuning do not re-spend quota.

10. Terms-of-service notes

Google lets **place IDs be stored indefinitely** but restricts storing other Places content beyond its terms; the monthly refresh from stored IDs and the rep's own notes are the practical mitigation, and Google asks for a "Source: Google" attribution when its data is shown outside a map. Data Axle via the library has its own terms for commercial use. Registry data (BOT, FSIS, ECHO, CalEPA, PRA) is public record. Not legal advice.

11. Where the code lives and how it runs

- `scripts/food_facilities/` in this repo: `sources/` (one module per registry), `places.py` (sweep + enrichment), `classify.py`, `sheet.py` (tab creation, formatting, upsert, other-tab checksum), own `requirements.txt`. **Not** added to the MCP server's runtime deps, Docker image or `requirements.lock`.
- Tests against recorded responses and a fake Sheets client; CI needs no keys.
- Env vars in `.env.example`: `GOOGLE_MAPS_API_KEY`, `GOOGLE_SHEETS_SA_JSON`, `LEADS_SHEET_ID`, `FOOD_TAB_NAME` (default `Food Facilities`), `OFFICE_LAT=38.6941`, `OFFICE_LNG=-121.4660`, `RADIUS_MILES=28`.
- Commands: `pull --dry-run` (CSV only), `pull --write` (first fill), `refresh` (monthly). Manual to start; Railway cron optional later.
- README section + CLAUDE.md pointer.

12. Phases

0. **Owner setup (blocks the rest).** Google Cloud project with billing; enable Places API (New) and Sheets API; API key restricted to Places; service account + JSON key; share the sheet with the service-account email as Editor; keys into `.env` /Railway only. Optional: send the CDPH PRA request I draft; get a Sacramento Public Library card for Data Axle. Answer section 13.
1. **Registry pull, dry run.** City BOT + FSIS + ECHO + CalEPA (+ CDFA) merged, filtered to 28 miles, classified, CSV out with counts by source/category/city. No Google key needed yet.

Owner spot-checks 30 rows.

2. **Google sweep + enrichment.** Keyword sweep, name+address lookups for registry rows, phones/websites/status, DNC and Leads cross-check, region labels. Second dry-run CSV.
3. **Tab creation and first write.** Create `Food Facilities`, formatting, dropdowns, write rows, verify other-tab checksums, hand to the rep.
4. **Refresh and docs.** `refresh` command, tests, README/CLAUDE.md/.env.example.
5. **Optional.** Merge the CDPH PRA list when it arrives; Data Axle export; FieldRoutes existing-customer flag.

13. Decisions needed before code

1. Column order A-Y as in section 6 (Name, Phone, Type, City, Notes, Follow-up date, ...)?
Alternative: Notes and Follow-up date as columns A-B. One-line change.
2. File the CDPH Public Records Act request for the Processed Food Registration list? (I draft, owner sends.)
3. Does anyone have a Sacramento Public Library card for the Data Axle export?
4. Include beverage producers (breweries, wineries, distilleries) and storefront bakeries, flagged for review?
5. Include institutional kitchens (school-district central kitchens, hospital food service, food bank)?
6. Keep rows already in `Leads` (flagged) or drop them?
7. Status dropdown values OK, or match what the rep uses in `Leads` ?
8. Cadence: one-time pull plus monthly refresh on request, or a Railway cron?
9. Who sets up the Google Cloud project and billing (Phase 0)?

14. Known limits

- Registries list what is permitted, not what is reachable: many rows will need Google or the rep for a phone.
- Google lists only mapped businesses with a profile; some plants show a corporate switchboard. Registries first is the fix for recall; Google is the fix for contactability.
- Categories are inferred; column T flags the uncertain ones and the first call fixes the rest.

Appendix: handoff inputs for whoever builds this

- Sheet ID `1u6e6psiW581rGr92r86aSJQW2XzPZfzsnhpJl0lB4F8`; tabs today: `Leads`, `Sig nals`, `Runs`, `DNC`; new tab `Food Facilities`.
- Office: 6948 West 2nd St, Rio Linda, CA 95673; 38.6941, -121.4660; radius 28.0 mi straight-line.
- City of Sacramento BOT: `https://services5.arcgis.com/54falWtcpty3V47Z/arcgis/rest/services/account_data_with_header_NEW/FeatureServer/0` (query with `where=1=1&outFields=*&f=json`, page with `resultOffset`).

- USDA FSIS MPI Directory: <https://www.fsis.usda.gov/inspection/establishments/meat-poultry-and-egg-product-inspection-directory> (CSV).
- EPA ECHO facility search: <https://echo.epa.gov/facilities/facility-search> (NAICS filter, CSV export).
- CalEPA Regulated Site Portal: <https://siteportal.calepa.ca.gov/nsite/> ; CalEPA Open Data for bulk downloads.
- CDPH Food and Drug Branch (PRA request): FDBfood@cdph.ca.gov, (916) 650-6500.
- Google: Places API (New) Text Search + Place Details; Sheets API v4 with a service account.